

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Fundamentals Of Data Science COURSE CODE: DSA04

COURSE OUTCOME COVERED

C03: Evaluate datasets using descriptive statistics, including summarization, outlier detection, and distribution analysis, and perform statistical inference through estimation and hypothesis testing. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 02 Total Marks:100 Annexure A

Data Interpretation

Code of Conduct:

I Anto Chriswin G / 192324093 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary) 5 Marks	Level 3 (Proficient) 3 / 4 Marks	Level 2 (Developing) 2 Marks	Level 1 (Beginning) 1 Mark	Marks
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data	
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	

Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	
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Annexure A

Problem 01: Hospital Patient Health Monitoring

Scenario

A hospital wants to investigate how patient lifestyle factors affect blood pressure levels. **Dataset**

Patient_ID	Daily_Exercise (Minutes)	BMI	Systolic_BP
P001	10	32	150
P002	15	31	148
P003	20	30	145
P004	25	29	142
P005	30	28	138
P006	35	27	134
P007	40	26	130
P008	45	25	126
P009	50	24	122
P010	55	23	118

Tasks

1. Find the mean and standard deviation of blood pressure.
2. Calculate covariance between:
 - o Exercise and Blood Pressure
 - o BMI and Blood Pressure
3. Calculate correlation coefficients.
4. Identify whether exercise or BMI is more strongly associated with blood pressure.
5. Discuss possible healthcare interventions.

Hospital Patient Health Monitoring Dataset

	Patient_ID	Daily_Exercise	BMI	Systolic_BP
0	P001	10	32	150
1	P002	15	31	148
2	P003	20	30	145
3	P004	25	29	142
4	P005	30	28	138
5	P006	35	27	134
6	P007	40	26	130
7	P008	45	25	126
8	P009	50	24	122
9	P010	55	23	118

Mean Blood Pressure = 135.3

Standard Deviation = 11.14

Covariance (Exercise, BP) = -168.06

Covariance (BMI, BP) = 33.61

Correlation (Exercise, BP) = -0.9969

Correlation (BMI, BP) = 0.9969

Exercise has a stronger association with Blood Pressure.

Healthcare Interventions:

- Encourage regular physical exercise.
- Maintain healthy body weight.
- Follow a balanced low-sodium diet.
- Schedule regular blood pressure monitoring.
- Provide lifestyle counseling and health education.

Problem 02: E-Commerce Customer Behaviour Analysis

Scenario

An e-commerce company wants to identify factors influencing customer spending. **Dataset**

Customer_ID	Website_Visits	Time_Spent (Minutes)	Purchase_Amount (₹)
C001	5	20	1200
C002	8	25	1800
C003	10	30	2500
C004	12	35	3200
C005	15	40	4000
C006	18	45	5000
C007	20	50	6200

C008	23	55	7300
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C009	25	60	8500
C010	28	70	10000

Tasks

1. Calculate mean, variance, and standard deviation of Purchase Amount.
2. Find covariance between:
 - o Website Visits and Purchase Amount
 - o Time Spent and Purchase Amount
3. Calculate correlation coefficients.
4. Determine which customer behavior metric better predicts spending.
5. Suggest marketing strategies based on the findings.

Customer_ID	Website_Visits	Time_Spent	Purchase_Amount
0	C001	5	20
1	C002	8	25
2	C003	10	30
3	C004	12	35
4	C005	15	40
5	C006	18	45
6	C007	20	50
7	C008	23	55
8	C009	25	60
9	C010	28	70

Mean Purchase Amount = ₹ 4970.0
Variance = 8793444.44
Standard Deviation = 2965.37

Covariance (Website Visits, Purchase Amount) = 22568.89
Covariance (Time Spent, Purchase Amount) = 47266.67

Correlation (Website Visits, Purchase Amount) = 0.9914
Correlation (Time Spent, Purchase Amount) = 0.9949

Time Spent better predicts customer spending.

Suggested Marketing Strategies:

- Improve website engagement to increase browsing time.
- Recommend personalized products.
- Offer discounts for frequent visitors.
- Introduce loyalty and reward programs.
- Optimize website content for better user experience.

Problem 03: Employee Productivity and Performance

Scenario

A software company is investigating the relationship between employee productivity and performance ratings.

Dataset

Employee_ID	Training_Hours	Projects_Completed	Performance_Rating
E001	5	2	60

E002	8	3	65
E003	10	4	70
E004	12	5	74
E005	15	6	79
E006	18	7	84
E007	20	8	88
E008	22	9	91
E009	25	10	95
E010	28	11	98

Tasks

1. Compute the mean, variance, and standard deviation of Performance Ratings.
2. Calculate covariance between:
 - o Training Hours and Performance Rating
 - o Projects Completed and Performance Rating
3. Compute correlation coefficients.
4. Identify the strongest factor affecting employee performance.
5. Recommend HR policies to improve employee productivity

```
Employee Productivity and Performance Dataset
Employee_ID Training_Hours Projects_Completed Performance_Rating
0 E001 5 2 60
1 E002 8 3 65
2 E003 10 4 70
3 E004 12 5 74
4 E005 15 6 79
5 E006 18 7 84
6 E007 20 8 88
7 E008 22 9 91
8 E009 25 10 95
9 E010 28 11 98
```

```
Mean Performance Rating = 80.4
Variance = 167.82
Standard Deviation = 12.95
```

```
Covariance (Training Hours, Performance Rating) = 97.87
Covariance (Projects Completed, Performance Rating) = 39.11
```

```
Correlation (Training Hours, Performance Rating) = 0.9957
Correlation (Projects Completed, Performance Rating) = 0.9972
```

Projects Completed is the strongest factor affecting performance.

Recommended HR Policies:

- Increase employee training programs.
- Encourage employees to complete more projects.
- Introduce performance-based rewards.
- Provide mentorship and skill development sessions.

Problem 04 : Smart City Traffic Analysis

A city administration wants to analyze traffic congestion patterns.

Day	Traffic_Volume (Vehicles)	Rainfall (mm)	Average_Speed (km/h)	Accident_Count
D1	12000	0	55	2
D2	13500	5	50	3
D3	14500	10	48	4
D4	16000	15	45	5
D5	17500	20	42	6
D6	19000	25	38	8
D7	21000	30	35	10
D8	22500	35	32	12
D9	24000	40	28	14
D10	26000	45	25	16

Tasks

1. Calculate descriptive statistics (Mean, Variance, Standard Deviation) for all variables.
2. Compute covariance matrices.
3. Compute correlation matrices.
4. Identify positive and negative relationships among variables.
5. Determine which factor is most associated with accident occurrence.
6. Provide data-driven recommendations to reduce traffic accidents.
7. Visualize relationships using scatter plots and interpret the results.

```
Smart_City_Traffic_Dataset
Day Traffic_Volume Rainfall Average_Speed Accident_Count
0 D1 12000 0 55 2
1 D2 13500 5 50 3
2 D3 14500 10 48 4
3 D4 16000 15 45 5
4 D5 17500 20 42 6
5 D6 19000 25 38 8
6 D7 21000 30 35 10
7 D8 22500 35 32 12
8 D9 24000 40 28 14
9 D10 26000 45 25 16
```

```
Mean
Traffic_Volume 18600.0
Rainfall 22.5
Average_Speed 39.8
Accident_Count 8.0
dtype: float64
```

```
Variance
Traffic_Volume 2.215556e+07
Rainfall 2.291667e+02
Average_Speed 9.773333e+01
Accident_Count 2.333333e+01
dtype: float64
```

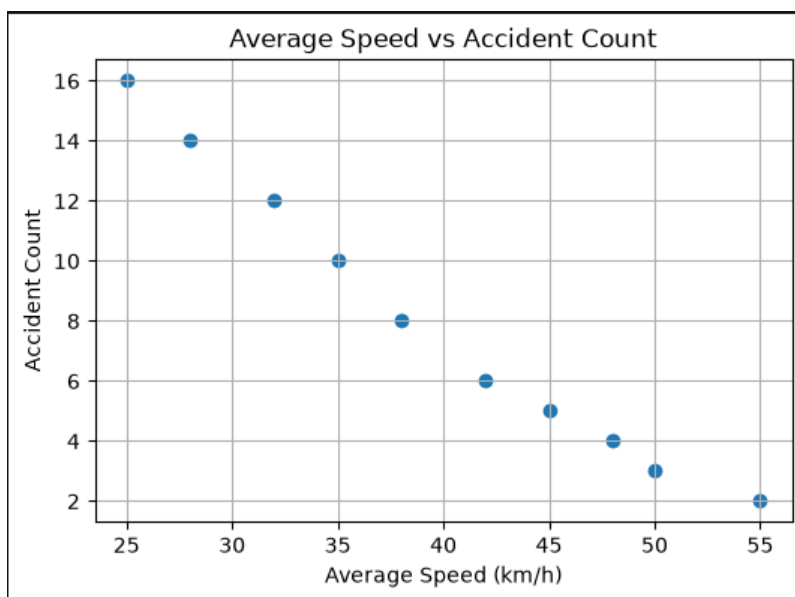
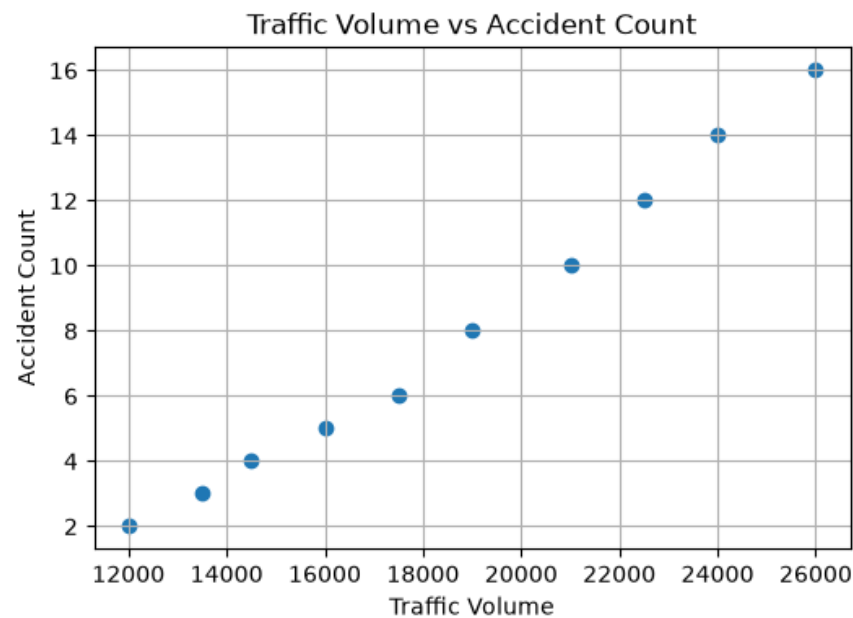
```
Standard Deviation
Traffic_Volume    4706.968829
Rainfall          15.138252
Average_Speed     9.886017
Accident_Count    4.830459
dtype: float64

Covariance Matrix
              Traffic_Volume  Rainfall  Average_Speed  Accident_Count
Traffic_Volume  2.215556e+07  71111.111111  -46422.222222  22611.111111
Rainfall        7.111111e+04   229.166667   -149.444444    72.222222
Average_Speed  -4.642222e+04  -149.444444    97.733333   -47.222222
Accident_Count  2.261111e+04   72.222222   -47.222222    23.333333

Correlation Matrix
              Traffic_Volume  Rainfall  Average_Speed  Accident_Count
Traffic_Volume    1.000000  0.997977   -0.997616    0.994471
Rainfall          0.997977  1.000000   -0.998580    0.987658
Average_Speed    -0.997616 -0.998580    1.000000   -0.988864
Accident_Count    0.994471  0.987658   -0.988864    1.000000

Relationship with Accident Count
Traffic_Volume    0.994471
Rainfall          0.987658
Average_Speed    -0.988864
Accident_Count    1.000000
Name: Accident_Count, dtype: float64

Most Associated Factor with Accident Count: Traffic_Volume
```



Problem 5: Average Delivery Time

A food delivery company records the delivery times (minutes) for 15

orders: 22, 25, 18, 30, 27, 24, 21, 19, 26, 23, 28, 20, 24, 25, 22

Tasks:

1. Find the point estimate of the population mean delivery time.
2. Find the point estimate of the population variance.
3. Interpret the estimates.

```
Average Delivery Time Analysis
Mean Delivery Time = 23.6 minutes
Variance = 11.4
Standard Deviation = 3.38 minutes
```

Problem 6: Customer Spending

A sample of 12 customers spent the following amounts (\$):

120, 145, 130, 155, 160, 125, 140, 135, 150, 170, 165, 155

Tasks:

1. Estimate the average spending.
2. Estimate the standard deviation.
3. Explain why these are point estimates.

```
Customer Spending Analysis
Average Spending = $ 145.83
Variance = 258.33
Standard Deviation = 16.07
```

Problem 7: Product Ratings

A company collects ratings from 20 customers:

4, 5, 3, 4, 5, 4, 3, 5, 4, 4, 5, 3, 4, 5, 4, 3, 5, 4, 5, 4

Tasks:

1. Compute the sample mean.
2. Compute the sample variance.
3. Compute the standard error of the mean.
4. Explain what the standard error indicates.


```
Product Ratings Analysis
Sample Mean = 4.15
Sample Variance = 0.5553
Sample Standard Deviation = 0.7452
Standard Error = 0.1666
```

Problem 8: Exam Scores

Scores of 25 students:

72, 75, 68, 80, 77, 73, 79, 81, 76, 74, 78, 69, 70, 82, 71, 75, 77, 73, 80, 76, 74, 79, 72,

81, 78 **Tasks:**

1. Calculate mean and standard deviation.
2. Calculate standard error.
3. Compare standard deviation and standard error.

```
Exam Score Analysis
Mean = 75.6
Standard Deviation = 3.96
Standard Error = 0.79
```

Problem 9: Manufacturing Weights

A sample of 50 packets has:

- Mean weight = 498 g
- Standard deviation = 12 g

Tasks:

1. Construct a 95% confidence interval for the population mean weight.
2. Interpret the interval.
3. Determine whether the target weight of 500 g is plausible.

```
Manufacturing Weights Analysis
Sample Mean = 498 g
Standard Error = 1.7 g
Margin of Error = 3.33 g
95% Confidence Interval = (494.67, 501.33) g
```

Problem 10: Customer Satisfaction

Two branches report:

Branch Mean Rating 95% CI

A 4.3 (4.1, 4.5)

B 4.0 (3.8, 4.2)

Tasks:

1. Compare the confidence intervals.
2. Can you conclude Branch A performs better?
3. Explain your reasoning.

```
Customer Satisfaction Analysis
-----
Branch A
Mean Rating : 4.3
95% Confidence Interval : (4.1, 4.5)

Branch B
Mean Rating : 4.0
95% Confidence Interval : (3.8, 4.2)

Comparison of Confidence Intervals
The confidence intervals overlap.

Conclusion
Cannot conclude that Branch A performs significantly better.
Reason: The 95% confidence intervals overlap,
so the observed difference may be due to sampling variation.
```

Problem 11: Average Battery Life

A battery manufacturer claims the average battery life is 10 hours. A sample of 36 batteries gives:

- Mean = 9.5 hours
- Standard deviation = 1.8 hours

Tasks:

1. Formulate null and alternative hypotheses.
2. Conduct a hypothesis test at $\alpha = 0.05$.
3. State your conclusion.

```
Null Hypothesis (H0):  $\mu = 10$ 
Alternative Hypothesis (H1):  $\mu < 10$ 
Z Statistic = -1.6667
P-value = 0.0478
Reject H0
```

Problem 12: Marketing Campaign

A company believes a new campaign increases

sales. Sales before campaign:

50, 52, 49, 55, 53, 51, 54, 50

Sales after campaign:

58, 60, 57, 62, 59, 61, 63, 58

Tasks:

1. Conduct a hypothesis test.
2. Calculate the p-value.
3. Interpret the p-value.
4. Make a decision at $\alpha = 0.05$.

```
T Statistic = 18.9315  
P-value = 0.0  
Reject H0
```

Problem 13: Call Centre Response Time

A sample of 100 calls has:

- Mean response time = 4.8 minutes
- Standard deviation = 1.5 minutes

Tasks:

1. Estimate the population mean.
2. Construct a 95% confidence interval.
3. Explain the Frequentist interpretation.

```
Estimated Population Mean = 4.8  
95% CI = (4.51, 5.09)
```

Problem 14: Defective Products

A sample of 500 products contains 22 defective items.

Tasks:

1. Estimate the defect rate.
2. Construct a 95% confidence interval.
3. Explain the Frequentist meaning of the interval.

```
import numpy as np

n = 500
x = 22
(variable) n: Literal[500]

p = x/n

se = np.sqrt((p*(1-p))/n)

lower = p - 1.96*se
upper = p + 1.96*se

print("Estimated Defect Rate =", round(p,4))
print("95% CI = ({:.4f}, {:.4f})".format(lower, upper))

Estimated Defect Rate = 0.044
95% CI = (0.0260, 0.0620)
```

Problem 15: Email Spam Detection

Historical information:

- Probability an email is spam = 0.25
- Probability spam email contains keyword "Offer" = 0.80
- Probability non-spam email contains keyword "Offer" = 0.10

Tasks:

1. Compute the probability that an email is spam given it contains "Offer."
2. Apply Bayes' theorem.
3. Interpret the result.

```
P_spam = 0.25
P_offer_given_spam = 0.80
P_offer_given_not = 0.10

P_offer = P_offer_given_spam*P_spam + P_offer_given_not*(1-P_spam)

posterior = (P_offer_given_spam*P_spam)/P_offer

print("Probability Email is Spam given Offer =", round(posterior,4))

Probability Email is Spam given Offer = 0.7273
```

Problem 16: Disease Diagnosis

Given:

- Disease prevalence = 2%

- Test sensitivity = 95%
- Test specificity = 90%

Tasks:

1. Compute the posterior probability of disease after a positive test.
2. Explain why the result differs from 95%.

```
prevalence = 0.02
sensitivity = 0.95
specificity = 0.90

false_positive = 1-specificity

positive = sensitivity*prevalence + false_positive*(1-prevalence)

posterior = (sensitivity*prevalence)/positive

print("Posterior Probability =", round(posterior,4))
```

Posterior Probability = 0.1624

Problem 17: Website Conversion Rate

A company tests two landing pages.

Group Visitors Purchases

A 5000 250

B 5200 312

Tasks:

1. Calculate conversion rates.
2. Formulate hypotheses.
3. Conduct a two-proportion z-test.
4. Determine whether page B performs better.

```
from statsmodels.stats.proportion import proportions_ztest

success = [250,312]
sample = [5000,5200]

z,p = proportions_ztest(success,sample,alternative='smaller')

rate_A = success[0]/sample[0]
rate_B = success[1]/sample[1]

print("Conversion Rate A =", round(rate_A,4))
print("Conversion Rate B =", round(rate_B,4))

print("Z =", round(z,4))
print("P-value =", round(p,6))

if p<0.05:
    print("Page B performs better")
else:
    print("No significant improvement")
```

Conversion Rate A = 0.05
 Conversion Rate B = 0.06
 Z = -2.2127
 P-value = 0.013459
 Page B performs better

Problem 18: Mobile App Feature Test

Feature A:

- 400 users
- Average engagement = 18 min
- SD = 5 min

Feature B:

- 420 users
- Average engagement = 20 min
- SD = 6 min

Tasks:

1. Test whether Feature B increases engagement.
2. Calculate test statistic and p-value.
3. Interpret results.

```
from scipy.stats import norm
import numpy as np

n1 = 400
mean1 = 18
sd1 = 5

n2 = 420
mean2 = 20
sd2 = 6

se = np.sqrt((sd1**2/n1)+(sd2**2/n2))

z = (mean2-mean1)/se

p = 1-norm.cdf(z)

print("Z =", round(z,4))
print("P-value =", round(p,6))

if p<0.05:
    print("Feature B increases engagement")
else:
    print("No significant increase")
```

```
Z = 5.195
P-value = 0.0
Feature B increases engagement
```

Problem 19: Recommendation System

A new recommendation algorithm increases click-through rate from 4.8% to

5.3%. Sample size:

- Old system: 100,000 users
- New system: 100,000 users

Tasks:

1. Test statistical significance.
2. Calculate p-value.
3. Discuss practical significance versus statistical significance.

```
from statsmodels.stats.proportion import proportions_ztest

old_users = 100000
new_users = 100000

old_clicks = int(0.048*old_users)
new_clicks = int(0.053*new_users)

z,p = proportions_ztest(
    [old_clicks,new_clicks],
    [old_users,new_users]
)

print("Z =", round(z,4))
print("P-value =", p)

if p<0.05:
    print("Statistically Significant")
else:
    print("Not Significant")
```

```
Z = -5.1058
P-value = 3.2943821279597856e-07
Statistically Significant
```

Problem 20: Salary Prediction Model

Two machine learning models produce Mean Absolute Errors (MAE): **Model MAE**

Model A 4200

Model B 4000

Across 50 test folds, the difference in MAE has:

- Mean difference = 200
- Standard deviation = 350

Tasks:

1. Perform a paired t-test.
2. Determine whether Model B significantly outperforms Model A.
3. Explain the result in terms of statistical significance.

```
from scipy.stats import ttest_1samp
import numpy as np

mean_diff = 200
sd = 350
n = 50

# Simulate differences using summary statistics
np.random.seed(1)
diff = np.random.normal(mean_diff, sd, n)

t,p = ttest_1samp(diff,0)

print("T Statistic =", round(t,4))
print("P-value =", round(p,6))

if p<0.05:
    print("Model B significantly outperforms Model A")
else:
    print("No significant difference")
```

```
T Statistic = 3.9413
P-value = 0.000257
Model B significantly outperforms Model A
```